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Teorema de diferenciación de Lebesgue en $\mathcal{L}^1$ para casi todo punto

Lema 1. *Sea $f \in \mathcal{L}^1(\mathbb{R})$ y $\lambda > 0$. Entonces, el conjunto*

$$A_\lambda := \left\{ x \in \mathbb{R} : \limsup_{r\to 0} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y)\,\mathrm{d}y - f(x) \right| > \lambda \right\}$$ *tiene medida de Lebesgue cero.*

***Demostración*:** Sea $\varepsilon > 0$. Como $\mathcal{C}_c(\mathbb{R})$ es denso en $L^1(\mathbb{R})$ (`teo-fn-continua-soporte-compacto-dens` $\exists g \in \mathcal{C}_c(\mathbb{R}) : \|f-g\|_1 \leqslant {}^{\varepsilon\lambda}\!/_8$. Como g es continua, por el `teo-diferenciacion-lebesgue-continuas/??` sabemos que

$$\forall x \in \mathbb{R} : \lim_{r\to 0^+} \frac{1}{2r} \int_{x-r}^{x+r} g(y)\,\mathrm{d}y = g(x).$$

Se tiene así que $\left| \frac{1}{2r} \int_{x-r}^{x+r} f(y)\,\mathrm{d}y - f(x) \right| =$

$$\begin{aligned} &= \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y)\,\mathrm{d}y - \frac{1}{2r} \int_{x-r}^{x+r} g(y)\,\mathrm{d}y + \frac{1}{2r} \int_{x-r}^{x+r} g(y)\,\mathrm{d}y - g(x) + g(x) - f(x) \right| \\ &\leqslant \frac{1}{2r} \int_{x-r}^{x+r} |f(y)-g(y)|\,\mathrm{d}y + \left| \frac{1}{2r} \int_{x-r}^{x+r} g(y)\,\mathrm{d}y - g(x) \right| + |g(x)-f(x)|\,. \end{aligned}$$

Como $\lim_{r\to 0^+} \frac{1}{2r} \int_{x-r}^{x+r} g(y)\,\mathrm{d}y = g(x)$, tomando el $\limsup$ cuando $r \to 0^+$, se tiene que

$$\begin{aligned} \limsup_{r\to 0} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y)\,\mathrm{d}y - f(x) \right| &\leqslant \limsup_{r\to 0} \frac{1}{2r} \int_{x-r}^{x+r} |f(y)-g(y)|\,\mathrm{d}y + 0 + |g(x)-f(x)| \\ &\leqslant M(f-g)(x) + |g(x)-f(x)|\,. \end{aligned} \tag{1}$$

Definimos ahora $B_\lambda := \left\{ x \in \mathbb{R} : M(f-g)(x) > \frac{\lambda}{2} \right\}$ y $C_\lambda := \left\{ x \in \mathbb{R} : |g(x)-f(x)| > \frac{\lambda}{2} \right\}$. Entonces, $A_\lambda \subset B_\lambda \cup C_\lambda$ porque $x \in (B_\lambda \cup C_\lambda)^\mathsf{c} = {B_\lambda}^\mathsf{c} \cap {C_\lambda}^\mathsf{c} \implies x \in {A_\lambda}^\mathsf{c}$ ya que

$$\begin{cases} x \in {B_\lambda}^\mathsf{c} \implies M(f-g)(x) \leqslant \lambda/2 \\ x \in {C_\lambda}^\mathsf{c} \implies |g(x)-f(x)| \leqslant \lambda/2 \end{cases} \implies \limsup_{r\to 0^+} \left| \frac{1}{2r} \int_{x-r}^{x+r} f(y)\,\mathrm{d}y - f(x) \right| \leqslant \lambda$$

por la desigualdad (**??**).

Por tanto, por subaditividad de la medida de Lebesgue, $m(A_\lambda) \leqslant m(B_\lambda) + m(C_\lambda)$. Ahora bien, por la desigualdad de Hardy-Littlewood, $m(B_\lambda) \leqslant \frac{3}{\lambda/2} \|f-g\|_1$. Entonces,

$$\begin{aligned} m(A_\lambda) &\leqslant \frac{3}{\lambda/2} \|f-g\|_1 + \int_{C_\lambda} 1 \, \mathrm{d}y \\ &\leqslant \frac{6}{\lambda} \|f-g\|_1 + \frac{2}{\lambda} \int_{C_\lambda} |g(x)-f(x)| \, \mathrm{d}x \leqslant \frac{8}{\lambda} \|f-g\|_1 \leqslant \varepsilon. \end{aligned}$$

Como $\varepsilon > 0$ era arbitrario, concluimos que $m(A_\lambda) = 0$. ∎

Referenciado en

- `teo-diferenciacion-lebesgue-l1`

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